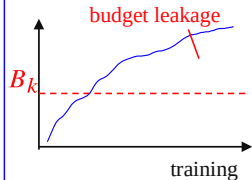


Penalty view

Costs enter the objective

$$A_r - \sum_k \lambda_k A_{c_k}$$

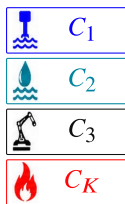


Fixed price can leave
a budget violated.
(schematic)

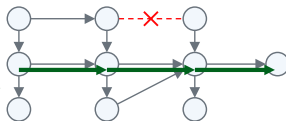
SafeOT

One flow prices all budgets

1. Costs



2. Flow solve



$$\langle C_k, F \rangle \leq b_k$$

flow balance · entropy

gray: observed · green: target

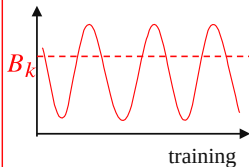
× hard event: excluded edge

can price before violation

Budget view

Budgets constrain costs

$$J_{c_k}(\pi) \leq B_k$$



Feedback-only prices
can react late and oscillate.
(schematic)

$\lambda_k^* \rightarrow \text{clip} + \text{EMA} \rightarrow$

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

observed-cost
feedback

$\rightarrow$ priced actor

PPO/TRPO $\rightarrow \pi_{t+1}$

One price rule

Fixed penalty

$c = 0, \beta$ fixed

exact settings

Lagrangian

$c = 0, \beta$ adaptive

SafeOT

finite ε

LP

$\varepsilon \rightarrow 0$

conditional limits

Active-set

$\varepsilon \rightarrow \infty$, saturated clip